

Data & AI Case Study

Mapped to two Wuerth internship roles

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This is my independent case study. I mapped the requirements of two Wuerth Data & AI internship postings to work I have already built and measured in my own portfolio (now 23 repositories, including the decision-chain integration capstone -- one real dataset through the whole distributor chain with 13 machine-checked reconciliation identities plus three additive ones -- an MCP agentic-integration server with contract-enforced result provenance, two warehouse-logistics flagships, a supply-network optimizer, an energy forecasting + dispatch study, a visual quality-inspection study with SPC monitoring, a fraud-operations study with gated retrain promotion, a predictive-maintenance policy study, and a live installable quantum-explainer PWA).

Job #1 (Agentic Automation with Low-code Platforms)
The use of AI workflows, Power BI, Power Automate, connecting systems & APIs, retail-analytics-real and decision-chain, measured and labelled on real public data) and exist to prove the methods work and are measurable -- not to forecast Wuerth outcomes. Agentic AI workflows, Power BI, Power Automate, connecting systems & APIs, document automation, ROI, rapid prototyping.

Job #2 -- Data & AI Analytics

BI / Power BI, KPI dashboards, forecasting & predictive analytics, Python & SQL, data modelling, turning data into decisions.

Who Wuerth is (public info)

Wuerth (public profile, approximate): a large family-owned German-HQ group in assembly/fastening and industrial MRO distribution -- ~400+ companies in 80+ countries, ~87,000 employees, ~EUR 20B+ revenue, a catalogue in the millions of articles, multi-channel (field sales, branches, e-commerce, e-procurement/EDI, ORSY inventory systems). A business that runs on assortment, pricing, availability, logistics, and high-volume transactional documents -- a natural fit for applied Data & AI.

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Opportunity map (summary)

Wuerth business area (public-info) -> Data/AI approach -> repo + measured result (synthetic unless noted)

Area (public-info)	Approach	Repo	Result (synthetic -- labelled if real)
Procurement / assortment / cross-sell	MILP vs greedy; Apriori/FP-growth; rule redundancy pruning	revops-optimizer, market-basket-analysis	Top hit 2.7x, 41% of rules redundant -- the 149 kept carry all the information; MASE 0.38; MAE-Policy move RMSE 0.948 (beat naive + Hoeffding) by 6% of the EUR17.4Bn working capital stated as a KPI even before the year-end review (exactly), MASE turns 0.369 after pacing fixed lead times obtain 8Q% inventory storage work check / loaded above the band, reported; rep decomposition (indirect standardization): 92% of the league table is territory, 8% execution
Pricing & margin	Elasticity + leakage; endogeneity fix; per-move robustness gate	revops-optimizer, sales-kpi-analytics, ml-models-lab	4.6% / 31% savings; 5% headroom: failing scenarios 96% -> 44%, expected day -4.6%
Inventory / replenishment	Newsvendor + forecast + ROP/EOQ policy + supplier reliability	distributor-intel-platform, ml-models-lab	≈EUR125k/yf modelled; 1,350 candidate slots v. ABN
Sales KPIs & analytics	KPIs + rolling-origin CV + pacing interval + rep decomposition	sales-kpi-analytics	ended at last contention win, far from cost model: state 0.354v0.347, improve 0.8% prettily sell the gate alone would lose money (modelled); the pre-mil carriage solver (modelled); engine slotting -44.2% (golden-zone 25% -> 100%) w/ 0.29% churn EOQ; Rolling>retail+200%vap eked €0.04m/bt, realized 0.028146% shorter; batching: savings -71.3%, routing flips to 12.2% cost
Logistics / routing	OR-Tools CP-SAT VRP + robustness	route-optimizer	stack 6.57% / 49.14 bids (ROI) winter's losses to 1750 NRE service point (peak); 97.9%; ±EUR11300/yort ASSUMED vs rf. battery does not pay back on these assumptions; backtest captures 72.7% of the LP bound (robust variant loses 7% reported)
E-procurement automation	Agentic + n8n low-code + seeded reliability benchmark	agentic-automation-lab	ROC AUC = inside 0.02 margin PUA recommended; Real ROI 6.7% vs 9.9% a 5% better abn and identification PAS 0.704v0.703; introduced; per-order spread to Champions 25.1% vs 69.0%; 165ydeleessy10%18ns 0.681mpis-sensit-evenus GBP 19,643,861.62 to the primary rows in many MYSC 3w82 bet gates -> MASE 3.782 vs 8.841 PROTECTOR 63
Document processing	Agentic extract + 7-rule validation gate + cost model	doc-extract-agent	PROTECTOR 45 new fraud, wins by shedding bid (stated); feed back 3m6 ranking saw veshe celebrating, we train ites + mps ova selector at the default threshold, reported; re-tuned economically the ranking flips to CBM (4.02) -- bounded, in-sample
Customer retention	Dartwin characterisation sim (Story Mode, 894-element plant, definable objects, multi-way line sim + fluids solver); slotting + DES + packing + pick-path routing + order batching	revops-optimizer, ml-models-lab	Censored Weibull + age-replacement vs condition-based + CBM re-tuning
Warehouse / WMS (new)	Logistics flow simulation (WarehouseTwin v3.19), pick-path routing + order batching	logistics-flow-studio (WarehouseTwin v3.19), logistics-digital-twin (engine)	
Supply-network design (new)	Facility MILP + flows + safety stock + service frontier + growth plan	supply-network-opt	
Energy management / facilities (new)	Load forecast (rolling CV) + peak-shaving LP + causal dispatch backtest	energy-demand-forecast	
Quality inspection (new)	Clean-only anomaly detection; pre-registered rule; SPC p-chart + WE rules	quality-anomaly-vision	
Real data (completed)	Cleaning + RFM + returns + lifecycle + leakage-safe CV	retail-analytics-real	
Chain integration & reconciliation (new)	Provenance-tagged pipeline + identity ledger; MCP agentic layer w/ machine-readable result provenance	decision-chain, chain-mcp	
Fraud & transaction-risk ops (new)	Cost-based alerting; gated retrain promotion; selective-labels feedback sim	fraud-detection-ops	
Maintenance & asset reliability (new)	Censored Weibull + age-replacement vs condition-based + CBM re-tuning	predictive-maintenance	

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Job #1 -- (Agentic) Automation with Low-code

Every posting requirement -> repo + artifact + measured proof (all synthetic data).

Requirement	Repo / artifact	Proof (synthetic unless noted)
Build AI-agent workflows -- and measure how they fail	agentic-automation-lab (+ eval/reliability.py benchmark)	agentic tool-use loops; reliability benchmark (1,350 seeded trials): content faults fail silently past the guards and retries can't zero them (assumed fault rates, stated); 51 tests 6 real engines as AI-callable tools; typed schemas; never-crash errors; contract-enforced machine-readable result provenance (data label + engine commit + determinism flag); 145 tests incl. live JSON-RPC handshake
Agentic integration via an open standard (MCP)	chain-mcp (official mcp Python SDK; Claude Desktop / Code configs)	
Low-code (n8n / Power Automate)	agentic-automation-lab: n8n/rfq_intake_agent.json	RFQ-intake agent; ~EUR625k/yr
Visual / flow-based building	agent-flow-studio	flow builder; ~EUR47k/yr
Connecting systems / APIs	agentic-automation-lab connector layer	agents call external APIs
Process automation (back-office)	doc-extract-agent	RFQ/invoice -> structured extract + validate; ~EUR145k/yr; combined-gate precision 70.0% -> 87.5%; cost model: auto-post pays only above 98.4% precision -- the gate alone would lose money (modeled), the pre-fill carries the value
Document intake / understanding	doc-extract-agent + 7-rule validation layer + cost model (eval/run_cost)	
Rapid prototyping	agent-flow-studio, agentic-automation-lab, logistics-flow-studio	runnable prototypes incl. a full installable offline WMS twin + factory/plant simulator (PWA, v3.19; 45 harnesses + 112/112 self-test)
Prototyping + education (hype-free)	quantum-explainer -- LIVE: dimitres-kisimov.github.io/quantum-explainer	hand-written simulator ~300 lines, zero deps; 42 physics assertions + 57 structural checks pass
Show automation ROI	automation-roi-explorer	EUR383k/yr net modelled
Python engineering	all automation repos	Python throughout
Stakeholder communication	automation-roi-explorer	business-readable ROI views

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Job #2 -- Data & AI Analytics

Every posting requirement -> repo + artifact + measured proof (synthetic data unless

Requirement	Repo / artifact	Proof of concept / initial results noted
One dataset through the whole chain (real data + labelled layers)	decision-chain (run artifact + offline dashboard; FAIL path per identity) revops-optimizer: powerbi/DAX_measures.md + kpi_uplift_risk	KPI layer + exec PDF; pacing EUR10.91M = 95.7% of plan, 80% interval; back-check: realised EUR11.28M landed above the band at honest misr rep decomposition under Monte-Carlo DAX back - 3 page report + Monte Carlo standardization; 96% of the leadline cable also open Is territory, torpedo tanks drivers; price-move gate: 17/29 ACCEPT, the biggest move HELD (67% carry) -- a screen, not a guarantee, bootstrap skill as churn W0.47yr capshn (cycle safety exactly), 5.5x turns, 99.9% fill; supplier reliability; measured lead times cost +14.4% safety stock -- wobble is 3/4 of the bill; frontier 6.3x convexity
BI / Power BI		MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
KPI dashboards / metrics + fair performance measurement	sales-kpi-analytics + pacing bullet chart + reperf.py revops-optimizer simulate.py + robustness.py; ml-models-lab BENCHMARK_CI.md	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Uncertainty / risk quantification	distributor-intelligence-platform dip/inventory.py + dip/reliability.py; supply-network-opt frontier	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Inventory policy / working capital		MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Predictive analytics / forecasting	sales-kpi-analytics; ml-models-lab global forecaster	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Forecasting rigour + reconciliation guard	distributor-intelligence-platform (MRO command center)	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Energy forecasting + optimization	energy-demand-forecast (forecasters + LP + dispatch backtest; 59 tests)	MASE 0.497 / MAPE 4.8%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918 MAPE < 4.6%, 14/14 folds MASE < 0.465m, 0.87d, RMSE 0.918
Python for analytics	sales-kpi-analytics, revops-optimizer	end-to-end pipelines
SQL / data modelling	sales-kpi-analytics SQL set	spend analysis queries
Excel-level tabular analysis	sales-kpi-analytics	spend breakdowns
Optimization / prescriptive	revops-optimizer; supply-network-opt; logistics-digital-twin	~EUR160k/yr; -21.2% cost vs greedy; ROC AUC 0.99; 2% cost savings +0.03; churn ECE 0.197 -> 0.021, bootstrap skill CI +0.457 [+0.381, +0.528]
Elasticity / statistical modelling	revops-optimizer; ml-models-lab	
Data into business decisions	sales-kpi-analytics; market-basket-analysis	AUR@1cm Package from crosssell 22c, bundle 4254 cases; affinity network > 92.0MB @ 3.6GB RAM was 8419 MBpm, 41% drop in peak memory overhead following load-shedding scenario initiated); spiceplan later feedback claim no impact new was increased and reduced by 2.8B x 2.73) better than previous time slot (AUC 0.92 vs prior which 0.84) tests down and fixed 1.08% test run; BOV test success 8.00% failed interday;
Market-basket / cross-sell	market-basket-analysis + affinity.py + redundancy.py	the model reported pre-launch > random; economically the ranking flip to CBM AT 0.006 AUC 0.579 vs baseline 0.72+- inside 0.006 AUC 0.579 vs baseline 0.72+- inside 0.006 AUC 0.579 vs baseline 0.72+- inside
Classification / anomaly detection	ml-models-lab; fraud-detection-ops	vs 0.393 to 0.15%; PPR: texture breaks hard to all resurrections exact optimum return calculation exact optimum optimized layout ~46% shorter; batching: savings -71.3%, routing flips 10x faster having 2; Robustness: 5%
Model lifecycle / MLOps (gated retrain)	fraud-detection-ops challenger.py + feedback.py	headroom cuts failures 96% -> 44%, expected day 0.6% safety stock 0.87M / 380 Lpe; reruns 9.6% growth plays 0.0% or value retained; Median To-day lag: champion AS 25% es 89.8%) lifecycle: resurrections outnumber repeats (439 vs 390/month); seasonal-naive wins CV, MASE 1.094
Reliability / maintenance (survival)	predictive-maintenance pdm/policy.py + pdm/cbm_tuning.py	
Visual quality inspection (vision)	quality-anomaly-vision (3 detectors, pre-registered rule; SPC; 45 tests)	
Warehouse / intralogistics + factory / process industry	logistics-flow-studio (WarehouseTwin v3.19); logistics-digital-twin (engine)	
Logistics / ops analytics	route-optimizer (36 tests); supply-network-opt (71 tests, growth.py)	
Real, messy data (completed)	retail-analytics-real (UCI Online Retail II, 1,067,371 real rows, CC BY 4.0)	
Present to decision-makers	revops-optimizer exec deck	exec narrative deck